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Aircraft Thrust Reverser Corrosion Shield

Abstract

A corrosion shield for an aircraft thrust reverser includes pivot door skins and side beam skins which cover the pivot doors and side beams of an aircraft thrust reverser. The pivot door skins and side beam skins may be secured using fasteners already included with the aircraft thrust reverser. The pivot door skins and the side beam skins are configured to protect surfaces of the aircraft thrust reverser from exhaust produced by an aircraft engine.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field

[0001] The disclosed embodiments relate generally to the field of aircraft thrust reversers. More specifically, the disclosed embodiments relate to the field of corrosion prevention on aircraft thrust reversers.

2. Description of the Related Art

[0002] Aircraft thrust reverser systems are known. For instance, U.S. Pat. No. 7,043,897 to Osman discloses a thrust reverser system having rectangular doors to optimize thrust reversal.

[0003] It is also known to form a thrust reverser system having a device for collecting cool air to cool the thrust reverser system. For instance, U.S. Pat. No. 10,605,196 to Caruel et al. discloses a device which collects cool air to cool the thrust reverser system and reduces the amount of Aluminum 2219 needed in the system.

[0004] It is also known to form a thrust reverser system from corrosion resistant materials. For instance, U.S. Patent Application Publication No. 2013/0009004 to Do discloses a thrust reverser system having components fabricated from a corrosion resistant material.

SUMMARY

[0005] This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Other aspects and advantages will be apparent from the following detailed description of the embodiments and the accompanying drawing figures.

[0006] In embodiments, an aircraft thrust reverser corrosion shield includes: a pivot door skin configured for mounting to a pivot door of an aircraft thrust reverser thereby protecting the pivot door from an exhaust of an aircraft engine; and a side beam skin configured for mounting to a side beam of the aircraft thrust reverser thereby protecting the side beam from the exhaust of the aircraft engine.

[0007] In some embodiments, an aircraft thrust reverser corrosion shield includes: a first pivot door skin configured for mounting to an inner surface of a first pivot door of an aircraft thrust reverser; a second pivot door skin configured for mounting to the inner surface of a second pivot door of the aircraft thrust reverser; a first side beam skin configured for mounting to the inner surface of a first side beam of the aircraft thrust reverser; and a second side beam skin configured for mounting to the inner surface of a second side beam of the aircraft thrust reverser, wherein the first and second pivot door skins and the first and second side beam skins each include an exhaust surface configured to face towards an exhaust nozzle of an aircraft engine.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0008] Illustrative embodiments are described in detail below with reference to the attached drawing figures, which are incorporated by reference herein and wherein:

[0009] FIG. **1**A is a perspective view of an aircraft thrust reverser installed onto an aircraft with thrust reverser doors closed;

[0010] FIG. **1**B is a perspective view of the aircraft thrust reverser of FIG. **1**A with thrust reverser doors opened;

[0011] FIG. **2**A is a perspective view of a pivot door of the aircraft thrust reverser of FIG. **1**A;

[0012] FIG. **2**B is a perspective view of a side beam of the aircraft thrust reverser of FIG. **1**A;

[0013] FIG. **2**C is a perspective view of the side beam and pivot door fastened to the aircraft thrust reverser of FIG. **1**A;

[0014] FIG. **3** is a perspective view of a corrosion shield configured for use with the aircraft thrust reverser of FIG. **1**A; and

[0015] FIG. **4** is a front view of the corrosion shield of FIG. **3** installed onto the aircraft thrust reverser of FIG. **1**A.

[0016] The drawing figures do not limit the invention to the specific embodiments disclosed and described herein. The drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the invention.

DETAILED DESCRIPTION

[0017] The following detailed description references the accompanying drawings that illustrate specific embodiments in which the invention can be practiced. The embodiments are intended to describe aspects of the invention in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments can be utilized and changes can be made without departing from the scope of the invention. The following detailed description is, therefore, not to be taken in a limiting sense. The scope of the invention is defined only by the appended claims, along with the full scope of equivalents to which such claims are entitled.

[0018] In this description, references to “one embodiment,” “an embodiment,” or “embodiments” mean that the feature or features being referred to are included in at least one embodiment of the technology. Separate references to “one embodiment,” “an embodiment,” or “embodiments” in this description do not necessarily refer to the same embodiment and are also not mutually exclusive unless so stated and/or except as will be readily apparent to those skilled in the art from the description. For example, a feature, structure, act, etc. described in one embodiment may also be included in other embodiments, but is not necessarily included. Thus, the technology can include a variety of combinations and/or integrations of the embodiments described herein.

[0019] Aircraft thrust reversers are positioned aft of an aircraft jet engine and provide an exhaust path for exhaust produced by the engine. Exhaust produced from an aircraft engine includes combustion byproducts that typically contain particle matter and is expelled at high temperatures which may corrode the surfaces of the aircraft thrust reversers exposed to the engine exhaust. In current arrangements, aircraft thrust reversers may be fabricated from aluminum and corrosion may be mitigated by applying a paint layer over the aluminum surface of the thrust reverser. The applied paint layer is often limited in its ability to withstand high temperatures and is therefore often not substantially effective in preventing corrosion to an aircraft thrust reverser.

[0020] Embodiments disclosed herein provide a corrosion shield for an aircraft thrust reverser. The corrosion shield for an aircraft thrust reverser includes a skin formed from an aluminum alloy which fastens onto the surface of the aircraft thrust reverser. The corrosion shield may be fastened using existing fasteners on the thrust reverser and may be adhered to the doors and side beams of the thrust reverser using an adhesive such that the corrosion shield is supported by the underlying structure of the thrust reverser. The corrosion shield is replaceable if the shield becomes damaged.

[0021] FIG. **1**A shows a perspective view of an aircraft thrust reverser **100** installed onto an aircraft **101**. The aircraft thrust reverser **100** has a circular or nearly circular exit and is positioned aft of the aircraft engine **106** such that exhaust expelled from an exit nozzle of the aircraft engine **106** is expelled through the thrust reverser **100**. In FIG. **1**A the aircraft thrust reverser **100** has its doors closed while in FIG. **1**B the doors of the aircraft thrust reverser **100** are open.

[0022] FIG. **2**A shows a pivot door **102**. The pivot door **102** is curved and gradually narrows from one end to the other to match the inner curvature of the aircraft thrust reverser **100**. The pivot door **102** may include structural holes **103** such that fasteners (e.g., bolts) may be inserted to secure the pivot door **102** to the aircraft thrust reverser **100**.

[0023] FIG. **2**B shows a side beam **104**. The side beam **104** is curved to match a portion of the inner curvature of the aircraft thrust reverser **100**. In embodiments, the side beam **104** includes outcropped portions and voids which may be strategically positioned to align with existing aircraft structure or fasteners of the thrust reverser **100**.

[0024] FIG. **2**C shows pivot doors **102** and side beams **104** installed onto the aircraft thrust reverser **100**. In FIG. **2**C, portions of the outer surfaces are depicted transparent for viewing underlying components. The curvature of the pivot doors **102** is substantially matched with the curvature of the aircraft thrust reverser **100** such that the pivot doors **102** when installed are substantially flush with an inner surface of the thrust reverser **100**. The pivot doors **102** may be positioned on upper and lower regions of the circular thrust reverser **100**. The side beams **104** are disposed on opposite sides of the aircraft thrust reverser **100** and in between the opposing pivot doors **102**. The curvature of the side beams **104** is also configured such that the side beams **104** are substantially flush with an inner surface of the thrust reverser **100** when installed. The aircraft thrust reverser **100** includes access panels or fasteners **108** which secure the pivot doors **102** and the side beams **104** which form the outer structure of the aircraft thrust reverser **100**. In embodiments, the fasteners **108** extend through the pivot doors **102** on either side of the side beams **104**. The pivot doors **102** include an exhaust surface **112** which is the surface exposed to the exhaust expelled by the engine **106**. The side beams **104** include an exhaust surface **114** which is the surface exposed to the exhaust expelled from the engine **106**. In some embodiments, the exhaust surfaces **112** and **114** may be coated with a paint having properties to slow/prevent corrosion. In some embodiments, the pivot doors **102** and/or the side beams **104** may include hinges or other pivotable mechanisms to allow the pivot doors **102** to pivot towards one another, decreasing the diameter of the aircraft thrust reverser **100** and redirecting a portion of the exhaust being expelled from the engine **106** (see FIG. **1**B). In embodiments, the pivot doors **102** and side beams **104** may be fabricated from an Aluminum alloy (e.g., Aluminum alloy 2219). The Aluminum alloy may be susceptible to corrosion when exhaust is expelled from the engine **106** and through the aircraft thrust reverser **100**.

[0025] FIG. **3** shows a perspective view of a corrosion shield **110**. In embodiments, the corrosion shield **110** includes pivot door skins **116** and side beam skins **118**. The pivot door skins **116** include an exhaust surface **122** and the side beam skins **118** include an exhaust surface **124** which are the inner facing surfaces of the corrosion shield **110** which face towards the center of the aircraft thrust reverser **100**. The pivot door skins **116** include a thrust reverser facing surface **126** and the side beam skins **118** include a thrust reverser facing surface **128** which are the outer facing surfaces of the corrosion shield **110**. The pivot door skins **116** are shaped to fit directly onto the exhaust surface **112** of the pivot doors **102** and the side beam skins **118** are shaped to be placed directly onto the exhaust surface **114** of the side beams **104** such that the thrust reverser facing surface **126** of the pivot door skins **116** directly contacts and is substantially flush with the exhaust surface **112** of the pivot doors **102**, and the thrust reverser facing surface **128** of the side beam skins **118** directly contacts and is substantially flush with the exhaust surface **114** of the side beams **104** (see FIG. **4**). The pivot door skins **116** include fastener holes **120** through the skin for the hinges. In some embodiments, the fastener holes **120** may have an irregular shape and are located such that the fasteners **108** of the pivot doors **102** may be used to secure the pivot door skins **116** to the pivot doors **102**. In other embodiments, the fastener holes **120** may be located on the side beam skins **118** or at other locations on the pivot door skins **116** such that fasteners **108** which secure the pivot doors **102** and side beams **104** to the aircraft thrust reverser **100** may be utilized to secure the pivot door skins **116** and side beam skins **118**. In some embodiments, dedicated fasteners may be added for securing the pivot door skins **116** and side beam skins **118** to the pivot doors **102** and side beams **104**. In some embodiments, a heat-resistant adhesive may be applied in addition to or in the place of fasteners to the exhaust surfaces **112** and **114** such that the pivot door skins **116** and side beam skins **118** may be adhered to the pivot doors **102** and side beams **104**. As best seen in FIG. **4**, the securement of the corrosion shield **110** directly onto the pivot doors **102** and side beams **104** substantially covers the exhaust surface **112** of the pivot doors **102** and exhaust surface **114** of the side beams **104** such that exhaust expelled from the engine **106** is directed onto the exhaust surface **122** and exhaust surface **124** of the pivot door skins **116** and side beam skins **118**, respectively, rather than the exhaust being directed onto the exhaust surfaces **112**, **114** of the thrust reverser **100**.

[0026] Advantageously, the pivot door skins **116** and side beam skins **118** of the corrosion shield **110** provide coverage and shielding of the exhaust surface **112** of the pivot doors **102** and exhaust surface **114** of the side beams **104** from expelled exhaust, which substantially reduces or prevents corrosion from occurring on the pivot doors **102** and side beams **104** of the thrust reverser **100**. In some embodiments, the corrosion shield **110** is fabricated from the Aluminum alloy 6000 and in other embodiments the corrosion shield **110** is fabricated from the Aluminum alloy 5000. The alloys Aluminum 6000 and Aluminum 5000 are each medium strength, corrosion resistant alloys. In other embodiments, other reasonably strong, corrosion resistant alloys could be used. In more specific embodiments, the Aluminum alloys 6061 and 5052 may be used which are both galvanically compatible with the underlying aluminum structure, which in embodiments are the pivot doors **102** and side beams **104**. The galvanic compatibility of the Aluminum alloys 6061 and 5052 and the underlying Aluminum structure allows for the contact between the exhaust surface **112** of the pivot doors **102** and the thrust reverser facing surface **126** of the pivot door skins **116**, and the contact between the exhaust surface **114** of the side beams **104** and the thrust reverser facing surface **128** of the side beam skins **118**, to not corrode each other from direct contact. In embodiments, the weight of the pivot door skins **116** is approximately one pound and in other embodiments the weight of the pivot door skins **116** may be approximately half to two pounds. In embodiments, the weight of the side beam skins **118** is approximately 0.25 pounds and in other embodiments the weight of the side beam skins **118** may be approximately 0.1 to 0.5 pounds. In embodiments, the corrosion shield **110** may be approximately two and a half pounds and in other embodiments the corrosion shield **110** may be approximately one and a half to three and a half pounds. In embodiments, the corrosion shield **110**, including the pivot door skins **116** and side beam skins **118**, is approximately 0.020 inches thick and in other embodiments the corrosion shield **110** may be 0.010 to 0.040 inches thick. The corrosion shield **110** is configured to last for the lifetime of the thrust reverser **100**.

[0027] Many different arrangements of the various components depicted, as well as components not shown, are possible without departing from the spirit and scope of what is claimed herein. Embodiments have been described with the intent to be illustrative rather than restrictive. Alternative embodiments will become apparent to those skilled in the art that do not depart from what is disclosed. A skilled artisan may develop alternative means of implementing the aforementioned improvements without departing from what is claimed.

[0028] It will be understood that certain features and subcombinations are of utility and may be employed without reference to other features and subcombinations and are contemplated within the scope of the claims. Not all steps listed in the various figures need be carried out in the specific order described.

Claims

1. An aircraft thrust reverser corrosion shield, comprising: a pivot door skin configured for mounting to a pivot door of an aircraft thrust reverser thereby protecting the pivot door from an exhaust of an aircraft engine; and a side beam skin configured for mounting to a side beam of the aircraft thrust reverser thereby protecting the side beam from the exhaust of the aircraft engine, wherein the pivot door skin and the side beam skin match up against each other, thereby substantially covering the pivot door and the side beam.

2. The corrosion shield of claim 1 wherein the pivot door skin and the side beam skin substantially cover an entire exhaust surface of the pivot door and the side beam, respectively.

3. The corrosion shield of claim 1 comprising a plurality of fasteners, wherein the fasteners are used to fasten the pivot door skin to the pivot door and the side beam skin to the side beam.

4. The corrosion shield of claim 3, wherein the fasteners comprise existing fasteners of the pivot door and the side beam.

5. The corrosion shield of claim 1 wherein the side beam skin and the pivot door skin each include at least one fastener hole, wherein each fastener hole provides access to a fastener.

6. The corrosion shield of claim 1 wherein the pivot door skin and side beam skin are each fabricated from an Aluminum 6000 alloy.

7. The corrosion shield of claim 1 wherein the pivot door skin and side beam skin are each fabricated from an Aluminum 5000 alloy.

8. The corrosion shield of claim 1 wherein the corrosion shield is 0.010 inch to 0.040 inch thick.

9. The corrosion shield of claim 1 wherein the pivot door skin and the side beam skin each comprise an outer surface configured for directly contacting the pivot door and an inner surface configured for exposure to the exhaust of the aircraft engine.

10. A corrosion shield for an aircraft thrust reverser, comprising: a first pivot door skin configured for mounting to an inner surface of a first pivot door of an aircraft thrust reverser; a second pivot door skin configured for mounting to an inner surface of a second pivot door of the aircraft thrust reverser; a first side beam skin configured for mounting to an inner surface of a first side beam of the aircraft thrust reverser; and a second side beam skin configured for mounting to an inner surface of a second side beam of the aircraft thrust reverser, wherein the first and second pivot door skins and the first and second side beam skins each comprise an exhaust surface configured to face towards a center of the aircraft thrust reverser of an aircraft engine, and wherein the first side beam skin abuts directly against a first edge of the first pivot door skin and abuts directly against a first edge of the second pivot door skin, and the second side beam skin abuts directly against a second edge of the first pivot door skin and abuts directly against a second edge of the second pivot door skin.

11. The corrosion shield of claim 10 wherein the first and second pivot door skins cover the pivot doors and the first and second side beam skins cover the side beams, thereby shielding the pivot doors and the side beams, respectively, from aircraft engine exhaust for mitigating corrosion of the pivot doors and the side beams.

12. The corrosion shield of claim 10 wherein the exhaust surface comprises a corrosion resistant material configured to withstand exposure to aircraft engine exhaust.

13. The corrosion shield of claim 12 wherein the corrosion resistant material comprises an Aluminum alloy.

14. The corrosion shield of claim 13 wherein the Aluminum alloy comprises Aluminum alloy 5052.

15. The corrosion shield of claim 13 wherein the Aluminum alloy comprises Aluminum alloy 6061.

16. The corrosion shield of claim 10 wherein the first and second side beam skins and the first and second pivot door skins each comprise a curvature that matches their respective pivot doors and side beams of the aircraft thrust reverser.

17. The corrosion shield of claim 10 wherein the first and second side beam skins and the first and second pivot door skins are each manufactured from an Aluminum alloy, wherein the Aluminum alloy is galvanically compatible between the pivot door skin and the pivot door and is galvanically compatible between the side beam skin and the side beam.